

# Integer Factorization Complexity

PAX L5 Narrow L2 General 27B | Anticloud FZ LLE

**Subject:** Proposed Hybrid Framework for Sub-Exponential Integer Factorization via Algebraic Number Theory and Quantum-Inspired Classical Optimization

## 1. Core Hypothesis

The hardness of integer factorization ( $N = p \times q$ ) does not stem solely from the combinatorial explosion of trial division, but from the inability to efficiently map the multiplicative group structure  $\mathbb{Z}_N^*$  onto a lattice with low enough density to permit polynomial-time sieving using classical resources.

**Hypothesis:** By constructing a generalized algebraic number field extension  $K(\sqrt{-D})$  where  $D$  is derived from the specific bit-pattern entropy of  $N$ , we can construct a "smoothness manifold" that allows for the reduction of the General Number Field Sieve (GNFS) complexity exponent from  $O(e^{\frac{64}{9} \frac{1}{3}} (\log N)^{\frac{1}{3}} (\log \log N)^{\frac{2}{3}})$  to  $O(e^{c \frac{1}{3}} (\log N)^{\frac{1}{3}} (\log \log N)^{\frac{1}{3}})$ , effectively bridging the gap between current classical limits and quantum polynomial time without requiring fault-tolerant quantum hardware. This relies on the conjecture that certain high-dimensional algebraic lattices associated with semi-primes exhibit hidden symmetries exploitable via classical randomized walks.

## 2. Mathematical Framework

Let  $N$  be an odd composite integer. We propose a modified Number Field Sieve (M-NFS) operating within a dual-field architecture:

**Field Construction:** Instead of standard irreducible polynomials  $f(x), g(x)$  used in GNFS, we define a dynamic ring extension  $\mathcal{O}_{K_N}$  over a quadratic imaginary field  $K = \mathbb{Q}(\sqrt{-\Delta_N})$ , where  $\Delta_N$  is selected based on the Jacobi symbol properties of  $N$ .

$$\Delta_N = \text{argmin}_d \left( \lambda_d(N) \right)$$

where  $\lambda_d(N)$  represents the estimated smoothness probability of elements generated by ideals in the ring  $\mathbb{Z}[\sqrt{-d}]$  modulo  $N$ .

**Lattice Mapping:** We map the logarithms of prime ideals in  $\mathcal{O}_{K_N}$  to a high-dimensional Euclidean space  $\mathbb{R}^k$  via the Minkowski embedding. The core innovation is the definition of a "Smoothness Potential Function"  $\Phi$ :

$$\Phi(v) = \sum_{i=1}^k w_i \log |v_i| - \epsilon \cdot \text{Entropy}(v)$$

Where  $w_i$  are weights derived from the distribution of small primes in the residue system of  $N$ .

**Optimization Objective:** The problem reduces to finding a non-trivial kernel element in the ideal class group  $\text{Cl}(K_N)$  such that the product of principal ideals yields a square relation:

$$\prod \mathfrak{p}_j^{e_j} = \alpha^2 \quad \text{and} \quad \prod \bar{\mathfrak{p}}_j^{e_j} = \beta^2$$

leading to  $x^2 \equiv y^2 \pmod{N}$  with  $x \not\equiv y \pmod{N}$ .

## 3. Proof Sketch / Key Steps

### Step 1: Entropy-Guided Polynomial Selection

Standard GNFS uses heuristic polynomial selection. We replace this with a deterministic search for polynomials minimizing the root-finding variance relative to the bit-depth of  $N$ .

**Sketch:** Define a cost function  $C(f) = \int |f(x)| dx$  weighted by the probability of  $f(x)$  being  $B$ -smooth. Prove that for large  $N$ , there exists a family of polynomials  $F_N$  where the average value magnitude scales as  $(\log N)^{1-\delta}$  rather than the standard exponential growth, reducing the sieve range required.

### Step 2: Accelerated Relation Collection via Random Walks

Utilize a Metropolis-Hastings algorithm adapted for the ideal class group.

**Sketch:** Construct a Markov Chain Monte Carlo (MCMC) process on the graph of ideals. The transition probabilities are biased towards ideals whose norms have high densities of small prime factors (smooth numbers).

\* **Key Lemma:** If the mixing time of this chain is bounded by  $\text{poly}(\log N)$ , then the collection of relations (pairs  $(a, b)$  such that  $a^2 \equiv b^2 \pmod{N}$ ) can be achieved in sub-exponential time. This depends on the spectral gap of the adjacency matrix of the ideal graph being inversely proportional to  $\log N$ .

**Step 3: Linear Algebra Reduction**

Once sufficient relations are gathered, solve the linear dependency problem over  $\mathbb{F}_2$ .

\* **Sketch:** Use sparse matrix Gaussian elimination (or Wiedemann's algorithm) optimized for the specific sparsity pattern induced by our entropy-weighted sieve. The dimension of the matrix will be reduced due to the higher yield of smooth relations per sieve step.

**Step 4: Square Root Extraction and GCD**

Compute  $x = \sqrt{\prod a}$  and  $y = \sqrt{\prod b}$  modulo  $N$ .

\* **Conclusion:** Calculate  $g = \gcd(x - y, N)$ . With high probability,  $1 < g < N$ , yielding a non-trivial factor.

### 4.